Prabhavit Gupta



Software Systems Engineer: Python, TypeScript, Automation & Cloud Services

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Summary

Software Systems Engineer with 3 years of commercial production experience architecting, implementing, and deploying high-availability software that integrates cloud backends with physical and operational automation pipelines. Strong background in Python, TypeScript, C++, and Linux environments, with demonstrated expertise building resilient microservices, diagnostic telemetry, and AI-assisted automation tooling. Ca' Foscari University of Venice alumnus based in Venice, Italy (CET time zone) with an EU Long-Term Residence Permit (permanent unrestricted EU work authorization); available immediately for full-time remote contract roles.

Technical Expertise

- **Software Development & Architecture:** Python (asyncio, FastAPI, PyTorch, NumPy, Pydantic), TypeScript / JavaScript, C++, low-level network and device communication protocols, distributed microservice patterns.
- **Cloud Infrastructure & DevOps:** AWS services, infrastructure-as-code automation (AWS CDK / CloudFormation), Docker containerization, CI/CD pipeline automation with GitHub Actions, Linux system internals.
- **Telemetry & System Reliability:** Distributed monitoring, Prometheus/Grafana metrics, automated alerting, diagnostic logging frameworks, and root-cause debugging under high operational loads.
- **Applied AI & Automation:** Tool-calling autonomous agents, Model Context Protocol (MCP), LLM-backed operational workflows, Claude Code for automated regression testing and high-velocity delivery.
- **Databases & Data Stores:** PostgreSQL (schema design, indexing, transactional integrity), Redis (caching, queues, distributed locks), structured telemetry storage.
- **Languages & Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Conversational). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Software & AI Automation Engineer *London, UK (remote)*

- Designed and operated scalable distributed software microservices in Python and TypeScript, supporting 4,500+ active enterprise users with 99.8% production reliability.
- Integrated cloud-based backend services with operational automation workflows, building event-driven message queues and reliable error recovery mechanisms.
- Implemented health-check monitoring, real-time alerting, and automated diagnostic tools across Linux containerized environments to ensure continuous operational uptime.
- Built AI-assisted automation pipelines using Claude and custom tool-calling agents, streamlining operational workflows and eliminating repetitive administrative tasks.
- Maintained strict continuous integration and automated testing routines with GitHub Actions and Playwright, preventing production regressions.

Urlink **Apr 2025 - Jul 2025**
Software Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Developed backend integration modules and API connectors linking third-party cloud services with distributed database clusters.
- Optimized data ingestion pipelines and database queries in PostgreSQL, eliminating concurrency bottlenecks and ensuring data consistency.
- Engineered automated system telemetry and error logging harnesses to monitor data pipeline health and flag anomalous spikes in real time.

Independent Technical Consultant **Aug 2023 - Feb 2024**
Software & Automation Developer *Venice, Italy*

- Built custom software tools, asynchronous data processing pipelines, and API integrations using Python and C++ for European SMB clients.
- Automated internal system workflows and device data reporting routines, reducing manual troubleshooting overhead

for technical client teams.
- Authored technical architecture specifications, deployment guides, and maintenance runbooks for Linux production systems.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous foundational coursework in formal mathematical logic, decision theory, analytical systems modeling, and epistemological frameworks.